Onboard Visual Foundation Models for Mars Terrain Perception and Rover Navigation

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Introduction

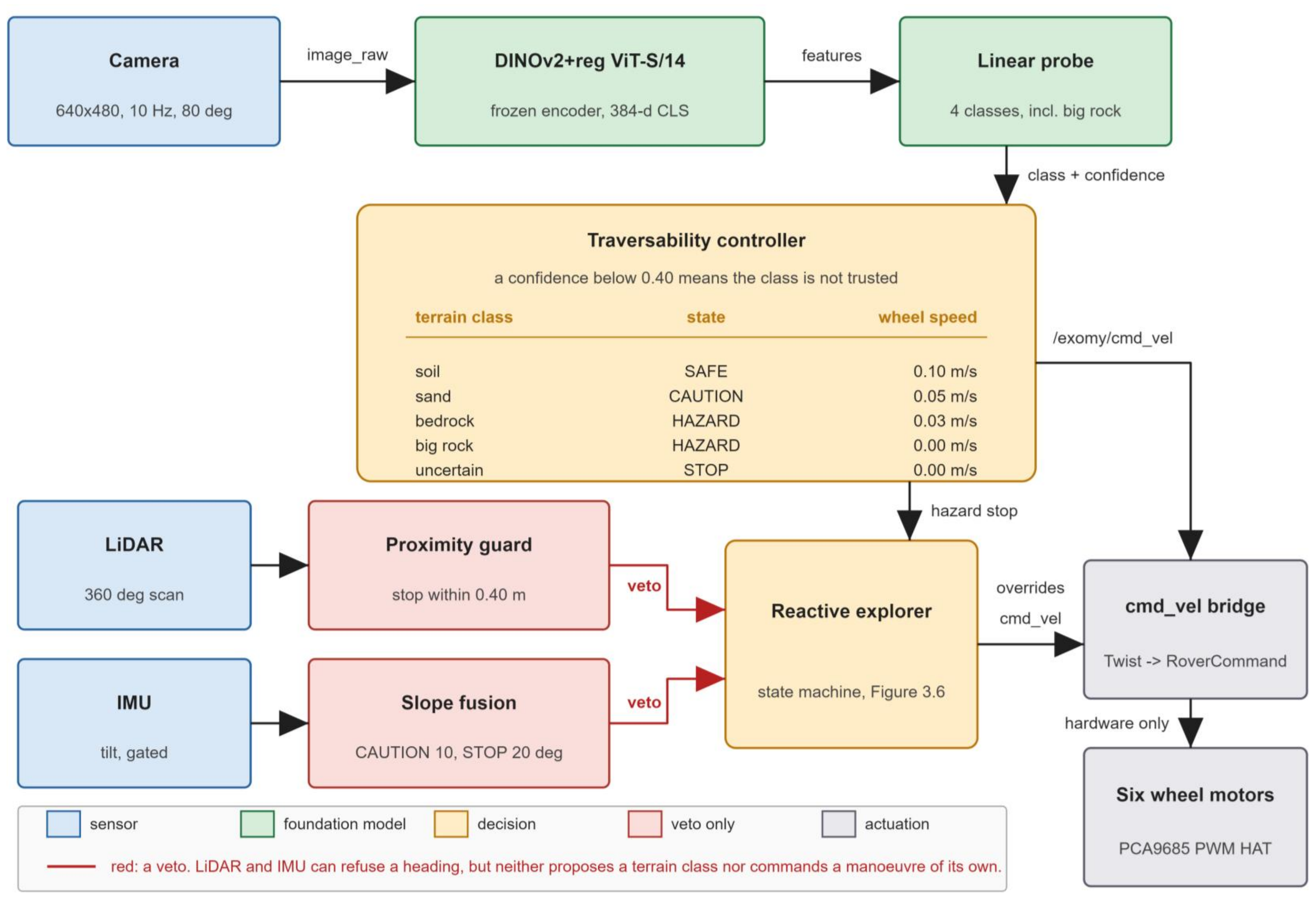
- A radio signal takes 3–22 minutes one way between Earth and Mars, so real-time control from Earth is impossible.
- Current rover hazard sensing relies heavily on geometry, but terrain type can still remain hidden.
- This work asks whether a frozen pretrained vision model can augment geometry-based safety with onboard visual terrain perception, and which pretraining objective transfers best to Mars terrain.

Aim and objectives

- Compare 22 frozen-encoder variants across eight pretraining objectives on AI4Mars.
- Deploy the strongest encoder suitable for a Raspberry Pi 4 within a ROS2 stack.
- Convert terrain class and confidence into conservative driving decisions, tested in Gazebo and on a real ExoMy rover.

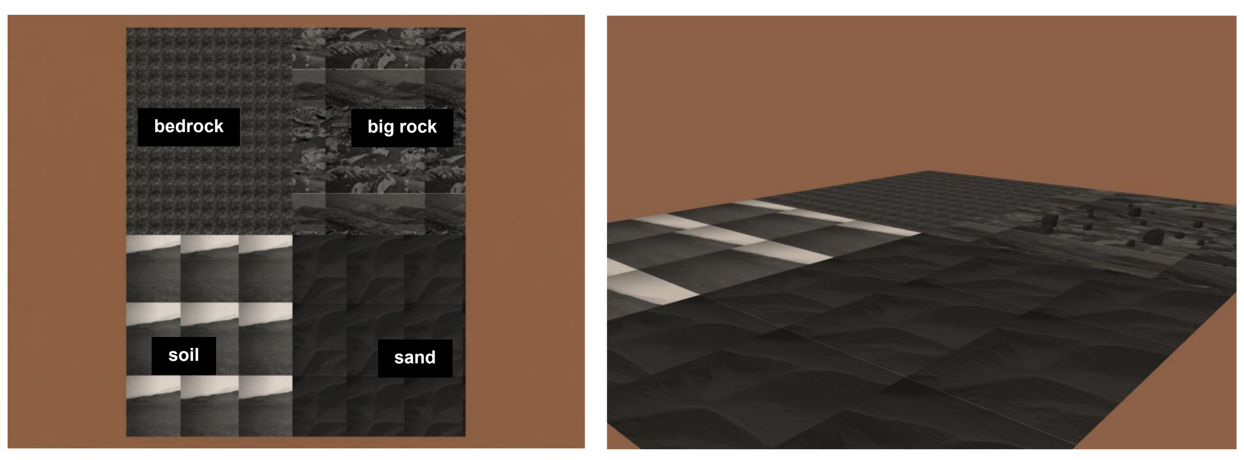
Method

- The DINOv2 encoder stays frozen; only a lightweight linear probe is trained on Mars labels.
- Terrain class and confidence determine the commanded action; confidence below 0.40 triggers STOP.
- LiDAR and IMU are vetoes only. Neither proposes a terrain class.



Camera to motion. The one trained component is the linear probe.

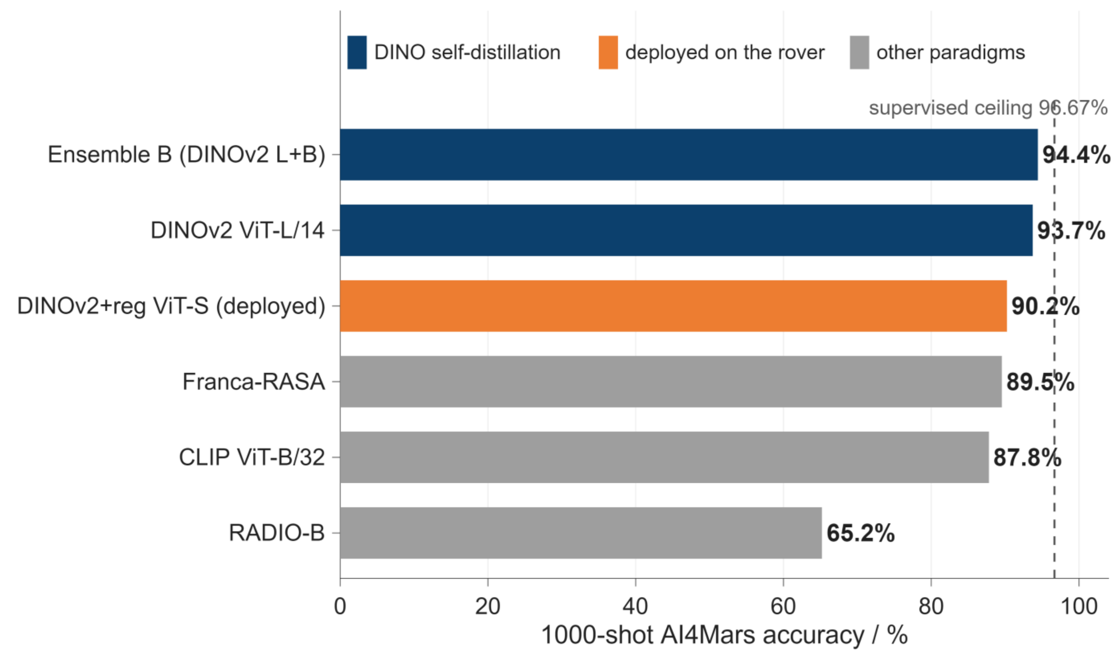
Simulation



A 30 × 24 m Gazebo world with four AI4Mars-textured quadrants, used to test the driving policy on known ground before it ever reached the rover.

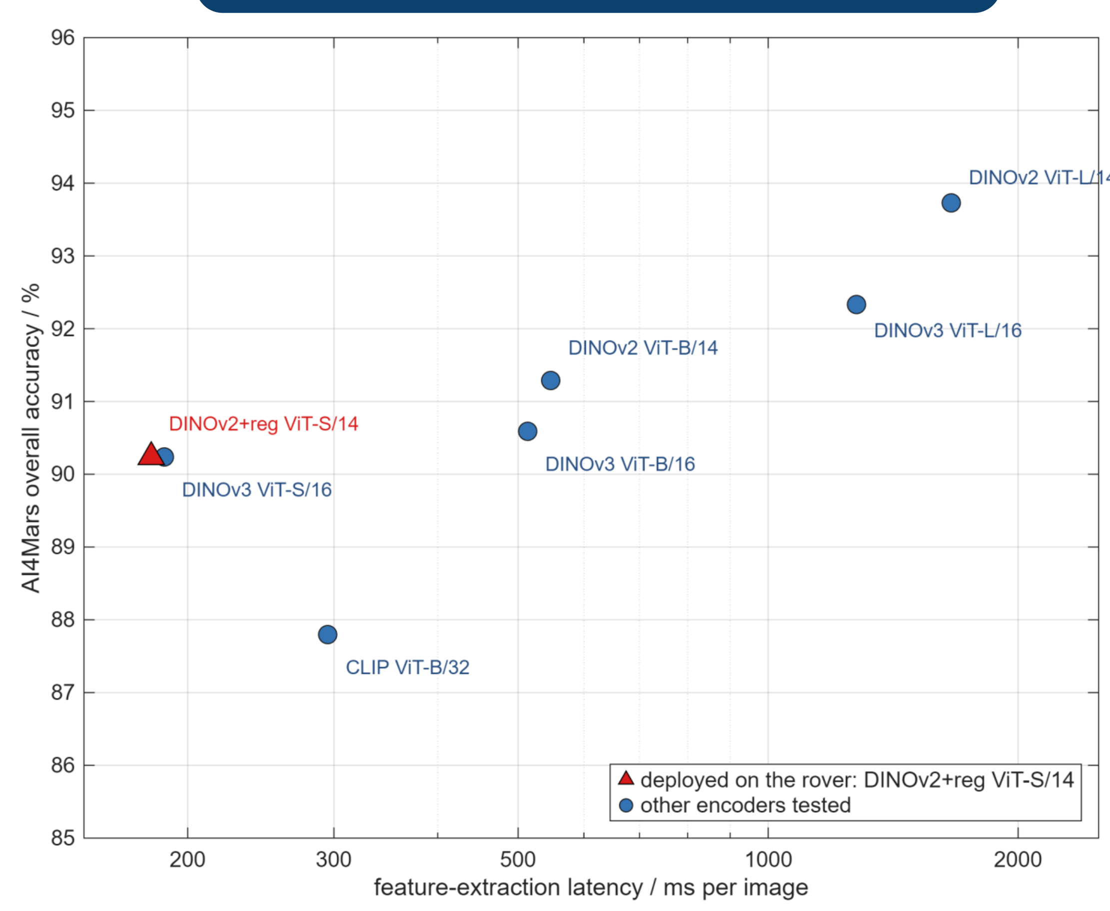
Model comparison

- DINO self-distillation gives the strongest overall results. A two-encoder DINOv2 ensemble reaches 94.43%, only 2.24 points below the supervised baseline.
- More pretraining data did not guarantee better transfer. DINOv2 outperformed DINOv3 at both ViT-B and ViT-L scales.



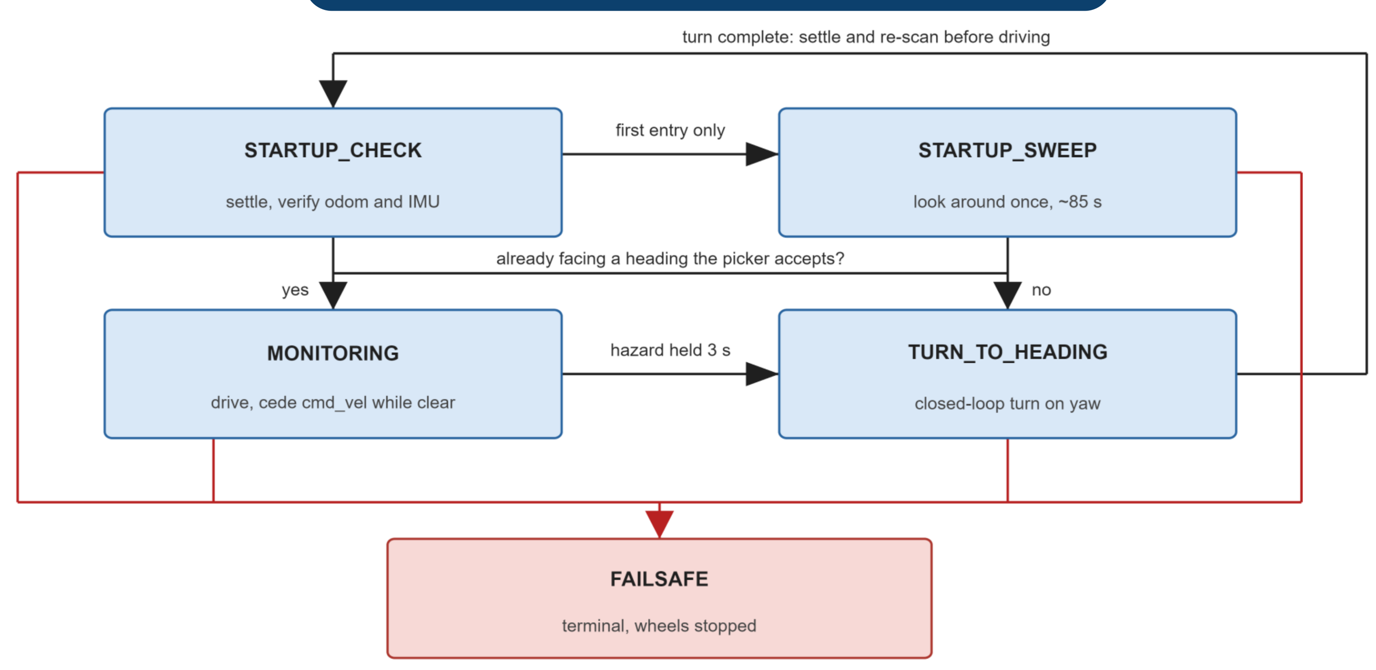
Six of the 22 model variants: the two best overall, the model deployed on the rover, and the strongest entry from each of three rival paradigms.

Deployment trade-off



DINOv2 ViT-L is 3.49 points more accurate, but feature extraction is about 9× slower. The smaller DINOv2+reg ViT-S/14 was deployed as the better onboard accuracy–latency trade-off.

Autonomy behaviour



The shipped state machine uses no reverse manoeuvre: without rear-facing sensing, unresolved hazards end in FAILSAFE and a full stop.

Results and conclusions

94.43%	90.24%	570 ms	3 / 5	5 / 5
best frozen AI4Mars accuracy	deployed-model AI4Mars accuracy	encoder only, Raspberry Pi 4	Gazebo classifications correct	Gazebo actions safe

1. **Frozen features are competitive.** A two-encoder DINOv2 ensemble reaches 94.43% on AI4Mars, only 2.24 points below the supervised baseline, while both encoders remain frozen.
2. **Conservative safety catches model errors.** The model classified 3/5 Gazebo positions correctly. Both rock zones were misclassified as sand, but a separate non-learned obstacle check stopped the rover, giving 5/5 safe actions.
3. **Real-camera transfer is class-dependent.** On rover-camera images, soil reached 100% and sand reached 73.9%, while big rock remained the main failure in real-camera transfer. Mean confidence was still 0.50, above the 0.40 stop threshold, showing the need for more big-rock data and better uncertainty awareness.

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